

Multimodal Diagnostics, Hazard Triage & Smart Artisan Matching

A comprehensive implementation blueprint enabling clients to describe faults via text, photos, or voice notes, receive instant root-cause diagnostics with cost estimates, and connect to high-fit verified artisans backed by escrow.

Stack: React 19 • Node.js Express 5 • PostgreSQL & Prisma • Gemini 2.0 Flash • Whisper

Target Markets: Lagos • Abuja • Port Harcourt

Status:
Implementation
Ready

1. Executive Summary & Problem Space

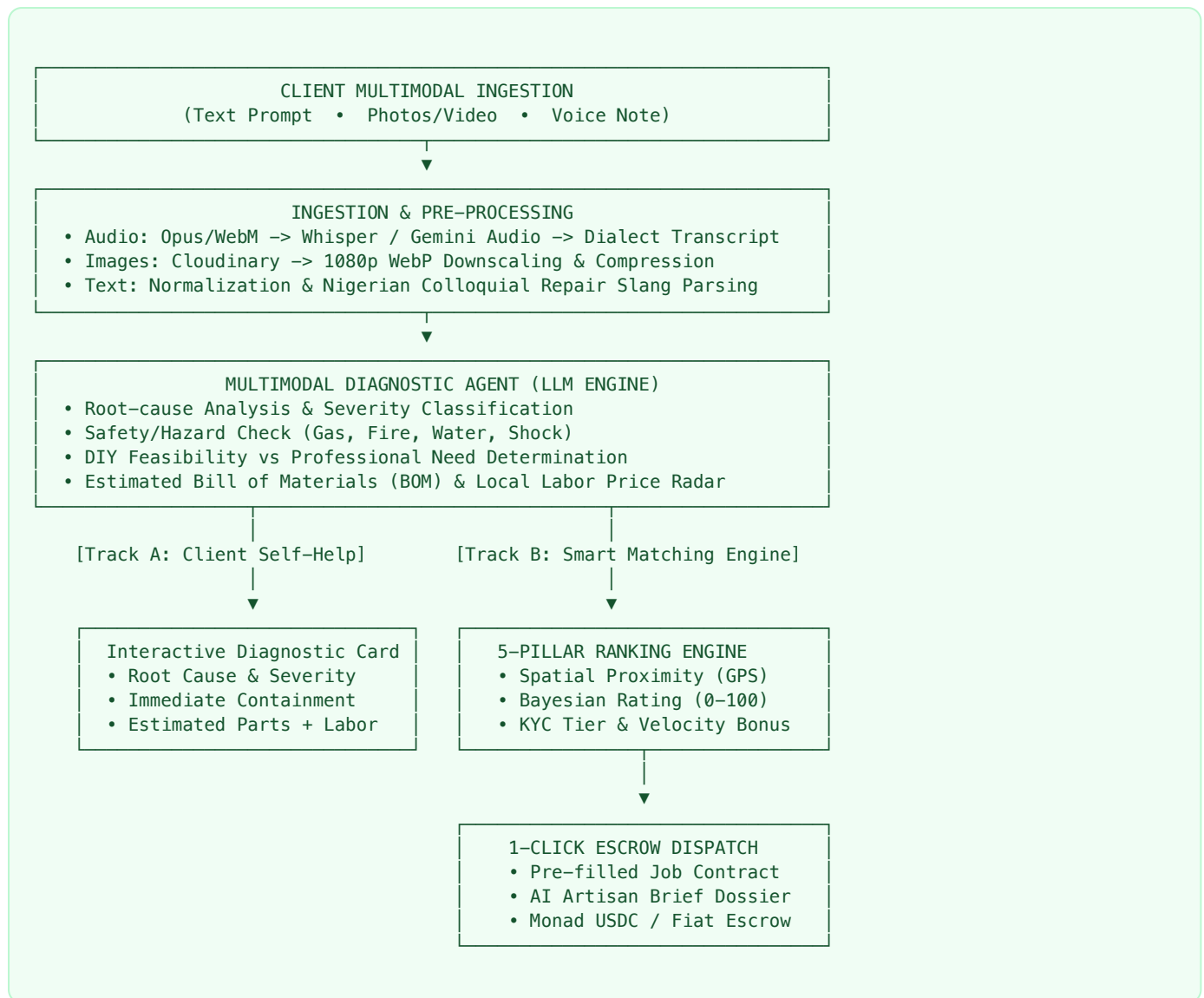
In trade services and home repairs, non-technical clients face acute barriers before an artisan is booked: lack of technical vocabulary, fear of overcharging, and frequent misdiagnoses leading to wrong artisan dispatches. The Artifix AI Copilot transforms Fixmate into an intelligent triage clinic.

The Triad of Repair Friction

- **Linguistic Deficit:** Homeowners describe symptoms ("ac dey sweat") rather than root causes ("blocked condensate drain").
- **Misdiagnosis:** Electrician called for compressor mechanics, resulting in wasted transit fees and delays.
- **Price Asymmetry:** Lack of objective insight into fair local labor rates and actual spare parts costs.

The Artifix Solution

- **Multimodal Ingestion:** Accepts spontaneous text, high-res photos, or 15–60s voice notes (in English or Nigerian Pidgin).
- **Dual-Track Output:** Immediate safety steps + price radar for clients, paired with automated job briefs for artisans.
- **5-Pillar Matching:** Ranked dispatch based on GPS distance, Bayesian rating, KYC, and portfolio similarity.



2. Multimodal Ingestion Pipeline

2.1 Voice Note Acoustic Architecture

Voice is the primary communication mode across West Africa. Clients record audio via the browser MediaRecorder API (compressed to 32kbps mono Opus in WebM containers). The audio buffer streams to Gemini 2.0 / Whisper primed with an extensive Nigerian trade terminology dictionary (e.g., *"pumping machine don knock"*, *"nepaa took light"*, *"gen dey surge"*, *"ac dey sweat"*).

2.2 Computer Vision Photo Pipeline

The client can attach up to 4 photos (wide shot, close-up of defect, serial/model plate, context). Before upload, the frontend compresses images to max 1920x1080 WebP. Cloudinary returns secure signed URLs which are fed directly into the Gemini multimodal vision endpoint.

2.3 Interactive Clarification Loop (Conversational Triage)

When the confidence score is below 70%, the AI pauses before matching and outputs 1–2 quick-tap multiple choice questions (e.g., *"Is the compressor fan spinning outside?"* [Yes] [No] [Cannot Access]).

3. Core Diagnostic Intelligence & Failure Analysis (FMEA)

Vertical	Common Symptoms	Probable Root Causes	Urgency	Immediate Safety Protocol
Plumbing	Under-sink leak, low shower pressure, gurgling drains	Worn spindle washer, calcified cartridge, pipe fissure	High	Isolate main stopcock; shut off booster pump
Electrical	Breaker tripping repeatedly, burning plastic smell	Overloaded circuit, burnt relay, shorted neutral wire	Critical	Trip main RCBO/Incomer breaker immediately; do not touch
HVAC / AC	Blower running lukewarm air, water dripping indoors	Failed run capacitor, low refrigerant, clogged tray	Moderate	Turn off AC from isolator switch to protect motor
Generators	Surging engine, cranking without ignition, black smoke	Fouled spark plug, dirty carburetor, bad AVR	Moderate	Turn off fuel petcock; disconnect load switch
Carpentry	Door sagging, jammed lock latch, sawdust on floor	Loose hinge screws, warped wood, termite infestation	Low	Avoid forcing handle; keep water away from timber

⚠ Real-Time Hazard Detection Protocol

When severity is marked as **CRITICAL** and `isHazardous === true` (active electrical arcing, gas smell, structural ceiling crack), the system immediately displays a high-visibility Emergency Alert Banner with step-by-step containment steps before any quotes or artisan matches are rendered.

4. Intelligent Artisan Matching Algorithm

Every candidate artisan in the target trade category is scored on a normalized 0 to 100 scale using the 5-pillar mathematical formulation:

Final Score Formula:

$$\text{Score}(a) = (0.35 \times \text{DistScore}) + (0.25 \times \text{RatingScore}) + (0.20 \times \text{PortfolioScore}) + (0.10 \times \text{KycScore}) + (0.10 \times \text{VelocityScore})$$

- Distance Score (35%):** Linear decay over a 20km radius using Haversine GPS distance. Artisans within 3km receive maximum points.
- Bayesian Rating (25%):** Normalizes ratings against review volume: $(v \times R + 5 \times 3.5) / (v + 5) \times 20$, preventing zero-review 5-star bias.
- Portfolio Similarity (20%):** Evaluates semantic match between diagnostic failure mode and verified portfolio tags.
- KYC Verification (10%):** 100 points for approved government identity verification (NIN/BVN), 0 points for unverified accounts.
- Completion Velocity (10%):** Reward for historical low dispute ratio and prompt contract turnaround.

5. Technical Implementation Blueprints

5.1 Database Models (Prisma Schema Additions)

```

model AiDiagnosticSession {
  id          String          @id @default(uuid()) @db.Uuid
  userId      String?         @map("user_id") @db.Uuid
  inputType   DiagnosticInputType @default(TEXT) @map("input_type")
  rawTextPrompt String?       @map("raw_text_prompt") @db.Text
  diagnosticTitle String?     @map("diagnostic_title") @db.VarChar(255)
  probableCause String?       @map("probable_cause") @db.Text
  severity    DiagnosticSeverity @default(LOW)
  urgency     DiagnosticUrgency  @default(ROUTINE)
  isHazardous Boolean          @default(false) @map("is_hazardous")
  safetyInstructions Json       @default("[]") @map("safety_instructions")
  diyAllowed   Boolean          @default(false) @map("diy_allowed")
  estimatedBOM Json            @default("[]") @map("estimated_bom")
  estimatedLaborMin Decimal?   @map("estimated_labor_min") @db.Decimal(12, 2)
  estimatedLaborMax Decimal?   @map("estimated_labor_max") @db.Decimal(12, 2)
  clientLatitude Decimal?      @map("client_latitude") @db.Decimal(10, 8)
  clientLongitude Decimal?     @map("client_longitude") @db.Decimal(11, 8)
  createdAt    DateTime        @default(now()) @map("created_at") @db.Timestamptz
  mediaFiles   DiagnosticMedia[]
  recommendations DiagnosticMatch[]
  @@map("ai_diagnostic_sessions")
}

```

6. Strategic Extra Features to Elevate the Platform

1. Artisan Prep Dossier & Pre-BOM

Generates an automated tool/part checklist so the artisan brings the correct capacitor or valve on Trip #1, eliminating the notorious 3-hour inspection market delay.

2. Anti-Gouging Fair Price Radar

Displays an objective median labor price band derived from Fixmate's historical transaction ledger for the client's specific LGA/City.

3. Two-Way Multimodal Audio Bridge

Converts client's formal problem statement into a 25s localized voice note in Nigerian Pidgin sent to the artisan's WhatsApp / PWA.

4. AI Post-Repair Before/After Verification

Compares pre-repair fault photos with post-repair completion photos before authorizing escrow milestone payout release.

5. Emergency 1-Tap SOS Dispatch

Bypasses normal 3-bid proposals when hazards are detected. Simultaneously pings the 3 closest emergency on-call artisans.

6. Interactive Visual AR Photo Annotation

Overlays glowing bounding boxes on the uploaded photo: *"Turn this blue valve 90 degrees clockwise to stop the leak."*

7. Implementation Roadmap & Phases

Phase	Timeline	Focus Area	Key Deliverables
Phase 1	Weeks 1 – 3	Core Multimodal Diagnostic MVP	Database migrations, Gemini 2.0 Flash JSON service, Cloudinary pipeline, React photo upload widget, basic category matching.
Phase 2	Weeks 4 – 6	Geospatial & Multi-Factor Matching	Haversine geospatial index, 5-pillar mathematical scoring, 1-click "Diagnosis to Job Draft" conversion, automated Artisan Dossier.
Phase 3	Weeks 7 – 9	Conversational Voice & Dialect Loop	Web Audio recorder with animated waveform, Whisper / Gemini Audio transcription, Nigerian Pidgin slang dictionary, TTS artisan briefings.
Phase 4	Weeks 10 – 12	Hardening, Escrow & Rollout	Direct Monad USDC & Paystack escrow linking, Emergency SOS dispatch, AI post-repair verification, Redis semantic caching, NDPR compliance.

8. SLA, Cost & Performance Matrix

Dimension	Target SLA	Engineering Strategy
Diagnostic Latency	< 2.5 seconds	Gemini 2.0 Flash with prompt caching; progressive status updates streamed to frontend.
Voice Transcription	< 1.8 seconds	Opus 32kbps client-side compression; direct buffer streaming to Whisper / Gemini.
AI Inference Cost	< \$0.004 / diagnosis	Strict max token budget (800 tokens); 1080p image downscaling; Redis symptom caching.
Matching Execution	< 350 ms	SQL bounding box pre-filtering before in-memory mathematical ranking calculation.
Fault Tolerance	99.9% Uptime	Graceful fallback to rule-based keyword matching and Fixmate live human triage queue.